

University of Nottingham Malaysia

SCHOOL OF COMPUTER SCIENCE
A LEVEL 1 MODULE, SUMMER SEMESTER 2021-2022
RESIT

COMPUTER FUNDAMENTALS (COMP1027)

Time allowed: **ONE** Hour

An additional 30 minutes to cover IT issues and upload time

Answer ALL questions

Submit your answers containing all the work you wish to have marked as a **single PDF file, with each page in the correct orientation, to the appropriate Dropbox on the module's Moodle page.**

Use the standard naming convention for your document: [Student ID]_[Module Code]. Write your student ID number at the top of each page of your answers. Do not include your name.

Although you may use any notes or resources you wish to help you complete this open-book examination, the academic misconduct policy that applies to your coursework also applies here. You must be careful to avoid plagiarism, collusion or false authorship. This statement refers to, and does not replace, the University policy which stipulates severe penalties for academic misconduct.

Staff are not permitted to answer assessment or teaching queries during the period in which your open-book examination is live. If you spot what you think may be an error on the exam paper, note this in your submission but answer the question as written.

IMPORTANT: Lateness penalties will not be applied, and late submission will not be accepted.

Marks available for sections of questions are shown in brackets in the right-hand margin

No calculators are permitted in this examination

1. Number Representation and Binary Arithmetic [18%]

- a. Given Table 1, calculate the 6 values labelled *a-f*. Show your calculation workings in detail and your representations for Binary, 1's Complement and 2's Complement must be in 8-bits width. (12)

Table 1.

Decimal	Binary	1's Complement	2's Complement
1111	<i>a</i>	<i>b</i>	<i>c</i>
<i>f</i>	<i>e</i>	<i>d</i>	0111 1111

- b. Given the following values of *x*, *y* and *z* below (in Binary):

- $x = 0110\ 1101$,
- $y = 1010\ 0110$,
- $z = 1000\ 0001$

Using **8-bits TWO's** complement, calculate the following the sums. Show your working in detail. Your answer should be in 8-bits width. (6)

- $x + y$
- $y - z$
- $x + z$

2. Boolean Algebra & Digital Logic [27%]

- a) Given the above **truth table** in Table 2:

Table 2.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

- i. Write out the expression (pre-simplified). (2)

P.T.O

Identity Name	AND Form	OR Form
Identity Law	$1x = x$	$0+x = x$
Null (or Dominance) Law	$0x = 0$	$1+x = 1$
Idempotent Law	$xx = x$	$x+x = x$
Inverse Law	$x\bar{x} = 0$	$x+\bar{x} = 1$
Commutative Law	$xy = yx$	$x+y = y+x$
Associative Law	$(xy)z = x(yz)$	$(x+y)+z = x+(y+z)$
Distributive Law	$x+yz = (x+y)(x+z)$	$x(y+z) = xy+xz$
Absorption Law	$x(x+y) = x$	$x+xy = x$
DeMorgan's Law	$(\bar{x}\bar{y}) = \overline{x+y}$	$(\overline{x+y}) = \bar{x}\bar{y}$
Double Complement Law	$\bar{\bar{x}} = x$	

Extracted from: L. Null, L. and J. Lobur, *The essentials of computer organisation and architecture*, Jones & Bartlett Publishers, 2003 (Fig. 1.3, P.26)

Figure 1.

- ii. Using the given Basic Identities of Boolean Algebra in Figure 1, simplify your expression in (a). Explicitly list the identity name that you use. Detail each step. (20)
- iii. Based on your answer in (b), draw the Logic Circuit Diagram to implement the circuit (using the relevant combinations of ONLY basic gates; AND, OR, NOT). (5)

3. Adders, MUX and ALU [30%]

- a. Illustrate the circuitry representation and structure (e.g., inputs and outputs) for a Half Adder and Full Adder. (11)
- b. For the ALU design (given in Figure 2), show (with *full working details*) how this ALU can achieve each of the listed output functionalities (*out*) according to the given corresponding control values in Table 3 (on page 4).

In your working details, you need to state what each of the variables ($x_1, x_2, y_1, y_2, K_1, K_2, Out_1, Out$) represents/contains in each case (a, b, c; in the table below): (19)

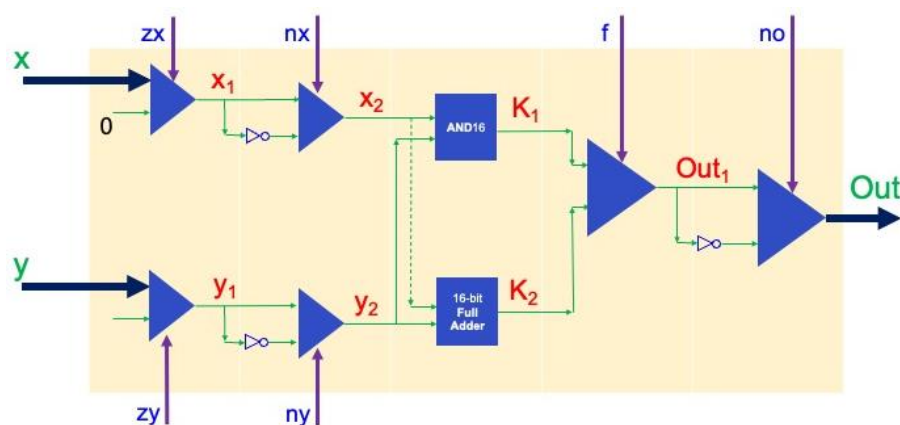


Figure 2.

P.T.O

Table 3.

	zx	nx	Zy	ny	f	no	Out
a.	1	0	1	0	1	0	0
b.	0	1	0	0	1	1	x - y
c.	0	0	0	1	1	1	y - x

Hint: $-A = A\sim + 1$ { where $A\sim$ is the bit-wise Not(A) }.

4. CPU Instructions and Networks [25%]

- a. The CPU, of the Hack Computer has three addressing modes; namely: **Immediate**, **Direct** and **Indirect** modes. Explain and compare between the 3 addressing modes. (10)

Table 4.

7	Application
6	-
5	-
4	-
3	-
2	Data Link
1	-

- b. Table 4 above shows the OSI Network stack, give the names for layers one, three, four, five and six. (5)
- c. Compare between the IPv4 and IPV6 addressing, in terms of format, number of possible addresses and size. (10)

END